Yinghui He

yh0068@princeton.edu | Website | Google Scholar |



Education

- **Princeton University** — *September 2024 – June 2028 (Anticipated)*
  Ph.D. Candidate in Computer Science. Advised by Sanjeev Arora. — Princeton, NJ
- **University of Michigan** — *September 2022 – May 2024*
  B.S.E. in Computer Science & B.S. (Minor) in Mathematics. (GPA: 3.98 / 4.00) — Ann Arbor, MI
- **Shanghai Jiao Tong University** — *September 2020 – August 2024*
  B.E. in Electrical and Computer Engineering. (GPA: 3.93 / 4.00; Ranking: 1 / 221) — Shanghai, China

Projects & Publications

PivotOPD: Learning to Recover from Pivotal Mistakes in Multi-Turn Agents — **Preprint (Sep 2026)**
Yinghui He, Yapei Chang, Khushi Bhardwaj, Daniele Molinari, Tugrul Konuk, Jan Kautz, and Ali Hatamizadeh. — *NVIDIA*

- Proposed PivotOPD, an on-policy distillation framework that identifies pivotal mistakes in multi-turn agents and jointly trains error prevention (reverse KL) and recovery (forward KL).
- Outperformed 13 baselines across ALFWorld, WebShop, and Search-based QA; improved ALFWorld success rate by 5.5% and SWE-Bench Verified resolve rate by 3.2%.

Toward Skill-Native LLMs: Skill Entropy for Benchmarking and Training Long-Horizon Reasoning — **NeurIPS 2026**
Yinghui He*, Ling Yang*, Jiarui Liu, Yongjin Yang, Lechen Zhang, Yingcheng Wu, Zhenfei Yin, Mengdi Wang, and Sanjeev Arora. — *Princeton University*

- Proposed Skill Entropy, a metric that quantifies the difficulty of transitions between reasoning skills in multi-step tasks.
- Released Skill^2-Bench (558 skills across 9 domains); developed Skill-Entropy RL that boosts smaller LMs on long-horizon reasoning.

Self-Distillation Zero: Self-Revision Turns Binary Rewards into Dense Supervision — **ICML 2026 RLxF Workshop (Best Paper); AI4Math Workshop (Honorable Mention)**
Yinghui He, Simran Kaur, Adithya Bhaskar, Yongjin Yang, Jiarui Liu, Narutatsu Ri, Liam Fowl, Abhishek Panigrahi, Danqi Chen, and Sanjeev Arora. — *Princeton University*

- Proposed SD-Zero, an on-policy self-distillation method that converts binary rewards into dense token supervision.
- Conducted experiments on Qwen3 and Olmo3 models across math and code benchmarks and brought 10% gains.

Skill-Targeted Adaptive Training — **ICLR 2026**
Yinghui He*, Abhishek Panigrahi*, Yong Lin, and Sanjeev Arora. — *Princeton University*

- Proposed STAT, an adaptive training strategy that constructs training data targeted at models' missing skills.
- Solely implemented data creation pipeline; conducted SFT and RL training on Llama and Qwen models.

AdaptMI: Adaptive Skill-based In-context Math Instruction for Small Language Models — **COLM 2025**
Yinghui He, Abhishek Panigrahi, Yong Lin, and Sanjeev Arora. — *Princeton University*

- Introduced AdaptMI, an adaptive method that boosts in-context performance by 6% through skill-based examples.
- Solely conducted experiments on Llama and Qwen models (1B–7B), and large-scale skill labeling with GPT-4o.

Hi-ToM: A Benchmark for Evaluating Higher-Order Theory of Mind Reasoning in LLMs — **EMNLP 2023**
Yinghui He*, Yufan Wu*, Yilin Jia, Rada Mihalcea, Yulong Chen, and Naihao Deng. — *University of Michigan*

- Released the Hi-ToM benchmark that tests the higher-order theory of mind abilities in LLMs (cited by 100+).
- Independently implemented an automated data generator; coordinated human annotation on 600+ samples.

LongProc: Benchmarking Long-Context Language Models on Long Procedural Generation — **COLM 2025**
Xi Ye, Fangcong Yin*, **Yinghui He***, Joie Zhang*, Howard Yen*, Tianyu Gao, Greg Durrett, and Danqi Chen. — *Princeton University*

- Introduced LongProc, a benchmark that tests both long-form generation and highly dispersed information.
- Independently designed and generated 2k examples for the *Theory-of-Mind Tracking* subset, and evaluated 17 LLMs.

EmoAgent: Assessing and Safeguarding Human-AI Interaction for Mental Health Safety — **EMNLP 2025 (Oral)**
Jiahao Qiu*, **Yinghui He***, Xinzhe Juan, Yimin Wang, Yuhan Liu, Zixin Yao, Yue Wu, Xun Jiang, Ling Yang, and Mengdi Wang. — *Princeton University*

- Built EmoAgent, a multi-agent AI framework to mitigate mental health risks by over 50% in human–AI interactions.
- Co-led experimental design and result analysis; coordinated the EmoAgent team.

Transferability for General Reasoning: An Automated Curriculum for Multi-Domain RLVR — **NeurIPS 2026; ICML 2026 RLxF Workshop (Best Paper)**
Yongjin Yang, Jiarui Liu, **Yinghui He**, Lechen Zhang, Bernhard Schölkopf, and Zhijing Jin. — *Princeton University*

- Proposed TAC, an online curriculum for multi-domain RLVR that models cross-domain transferability.
- Estimated transferability via projected gradient alignment (<1% overhead); up to +2.8 points on six domains.

Rethinking On-Policy Self-Distillation for Thinking Models — **NeurIPS 2026**
Simran Kaur, Narutatsu Ri, **Yinghui He**, Liam Fowl, and Sanjeev Arora. — *Princeton University*

- Showed privileged-context self-distillation hurts thinking models, with up to 17% relative drops on AIME/HMMT.
- Traced the failure to high-entropy forking tokens, where privileged context suppresses verification and backtracking.

Internship

- **NVIDIA** — *May 2026 – August 2026*
  Research Scientist Intern — Santa Clara, CA
  - Involved in the development of NVIDIA Nemotron models.
  - Studied algorithms & data curation for on-policy distillation on multi-turn agentic settings.
- **VMware** — *July 2022 – October 2022*
  Research Intern — Shanghai, China
  - Studied algorithms and applications of federated analytics; proposed optimization methods on federated analytics.

Honors and Awards

- **Best Paper Award, ICML 2026 RLxF Workshop** — *July 2026*
  Self-Distillation Zero: Self-Revision Turns Binary Rewards into Dense Supervision (first author)
- **Best Paper Award, ICML 2026 RLxF Workshop** — *July 2026*
  Transferability for General Reasoning: An Automated Curriculum for Multi-Domain RLVR
- **Honorable Mention, ICML 2026 AI4Math Workshop** — *July 2026*
  Self-Distillation Zero: Self-Revision Turns Binary Rewards into Dense Supervision (first author)
- **Mathematical Contest in Modelling** — *Finalist Award (2%), May 2022*
  Consortium for Mathematics and its Applications

Scholarships

- **Chinese National Scholarship**, Ministry of Education of the People's Republic of China — *¥8000 ($1170)*
  - Highest scholarship in China awarded to Top 1% college students.
- **Fan Xuji Scholarship**, Shanghai Jiao Tong University — *¥20000 ($2940)*
  - Highest SJTU scholarship awarded to 18 all-round developed students.
- **SJTU Undergraduate Excellence Scholarship**, Shanghai Jiao Tong University — *¥1000 ($150)*
  - Awarded to top-ranking (10%) students.
- **John Wu and Jane Sun Excellence Scholarship**, UM-SJTU Joint Institute — *¥75000 ($11000)*
  - Highest scholarship in UM-SJTU Joint Institute awarded to 10 all-round developed students.
- **The Jackson and Muriel Lum Scholarship**, UM-SJTU Joint Institute — *$14000*
  - Awarded to 5 excellent students in the dual-degree program.

Teaching & Advising

- **Teaching Assistant** — *September 2021 – August 2022*
  Physics II (60 students), Chemistry I (150 students) — Shanghai Jiao Tong University
- **Student Advisor** — *September 2021 – August 2022*
  UM-SJTU Joint Institute Advising Center — Shanghai Jiao Tong University
- **Writing Consultant** — *September 2021 – August 2023*
  UM-SJTU Joint Institute Writing Center — Shanghai Jiao Tong University